Python for computational Problem Solving · Unit_01 · Unit 1: Introduction

DRAFT — UE26CS151A · Unit_01 · Unit 1: Introduction · Process of Computation Problem Solving - Analysis, Design, Implementation, Testing

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Process of Computation Problem Solving - Analysis, Design, Implementation, Testing

1 Process of Computation Problem Solving - Analysis, Design, Implementation, Testing

Study focus: Process of Computation Problem Solving - Analysis, Design, Implementation, Testing. The notes explain the key terms, distinctions, and practical method used in this lesson. Read the cited transcript intervals for source context, then use the examples to check your understanding. EV0000034 · EV0000035

SHORT-NOTE RECAP

Separate the stages of solving a computational problem. Analysis identifies the required result, inputs, constraints, and special cases. Design turns that understanding into an algorithm or data representation. Implementation expresses the design in a programming language.

Testing compares actual outputs with expected behavior on representative and boundary inputs. A syntax error means the program violates Python's grammar; a logic error means it runs but computes the wrong result. Passing one example is not proof of correctness, so use cases that challenge assumptions.

There may be several valid designs. Compare them against clarity, correctness, and resource requirements instead of treating the first working program as the only solution.

EV0000034 · EV0000035 · EV0000036 · +17 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Process of Computation Problem Solving - Analysis, Design, Implementation, Testing transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_01/AV_Summaries/003_Process_of_Computation_Problem_Solving_-_Analysis_Design_Implementation_Testing__c5451c431739faf6.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Process of Computation Problem Solving - Analysis, Design, Implementation, Testing transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_01/AV_Summaries/003_Process_of_Computation_Problem_Solving_-_Analysis_Design_Implementation_Testing__c5451c431739faf6.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Process of Computation Problem Solving - Analysis, Design, Implementation, Testing transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_01/AV_Summaries/003_Process_of_Computation_Problem_Solving_-_Analysis_Design_Implementation_Testing__c5451c431739faf6.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9986; only the JSON segmentation supplies the displayed timestamps.
Process of Computation Problem Solving - Analysis, Design, Implementation, Testing transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_01/AV_Summaries/003_Process_of_Computation_Problem_Solving_-_Analysis_Design_Implementation_Testing__c5451c431739faf6.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.

2. 16 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
3. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
4. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
5. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
6. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00014	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0046. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0047 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00015	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0046. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0047 to s0063. Sidecar variants compare...
SRC00016	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0046. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0047 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00017	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0046. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0047 to s0063. Sidecar variants compare...

C Revision and retrieval practice